

Failure Points and How to Fix Them

There is a simple and elegant equation that is traditionally used to determine running velocity in endurance events. It states that velocity is determined by how big your aerobic engine is (VO₂ max), the fraction of it that you are utilizing (F), and the oxygen cost it takes for you to run a given distance (cost of running [Cr]). (These three variables should be familiar, since we covered them in the previous chapter.) The equation explains why a runner with a bigger aerobic engine can run faster than one with a smaller engine when they are equally economical. It also explains why the cost of running is important, particularly if you have maximized your aerobic engine. Consequently, innumerable hours of scientific research have been dedicated to studying how these three simple variables are affected by body weight, temperature, shoe mass, genetics, elastic energy return, flexibility, running cadence, stride length, training, gender, age, biomechanics, and many other factors.

$$Velocity = F \times VO_2max / Cr$$

The equation is extremely versatile, allowing us to predict and explain performance in endurance events from 5Ks to marathons (Di Prampero 1992). Furthermore, improving performance for running events ranging from the 5K to the marathon has been rooted in optimizing these three variables. Improve your VO₂ max and the fraction that you can utilize, and you will run faster. Reduce your cost of running, and you can run faster still.

But beyond the marathon distance, the validity of this equation in predicting performance starts to break down. Ultrarunning is not a marathon. It's not even a long marathon. It's a different sport altogether, and the variables associated with it mean that performance can no longer be determined solely by oxygen consumption and the cost of running.

The Science of Ultrarunning

While other endurance sports such as triathlon, marathon, and cycling have an abundance of scientific research to draw upon, ultrarunning has very little by comparison. It's a difficult proposition for a researcher. Finding subjects who are willing to run on a treadmill for the necessary durations is understandably challenging. And because of the remote nature of most ultras, fieldwork and race-day biological assessments are difficult to attain. Plus, taking post-race measurements involves poking and prodding athletes who in many cases have just finished the most difficult race of their lives. With all these obstacles, the scope of what can be studied will always be somewhat limited.

When I first started working with ultrarunners in the early 2000s, I followed a similar framework for any new sport group that I worked with: study the game, understand the athlete, and do research on the sport. The first two I could accomplish on my own. I could look at race profiles, dig into race reports, attend events, and even do races myself to better study ultrarunning as a sport. Then, the process of understanding the athlete was universal across

runners, cyclists, and triathletes, so I had that down pat, too. However, when it came to finding research in the early 2000s, I was grasping at straws. There was quite simply very little research to draw upon that was specific to ultramarathon running. Fortunately, that has changed. Ultrarunning's popularity and interest in the sport continue to grow, and with that comes academic research specific to the sport. The amount of research is still dwarfed by what has been published on the traditional sports of road and track running, cross-country skiing, cycling, and triathlon, but it's a start.

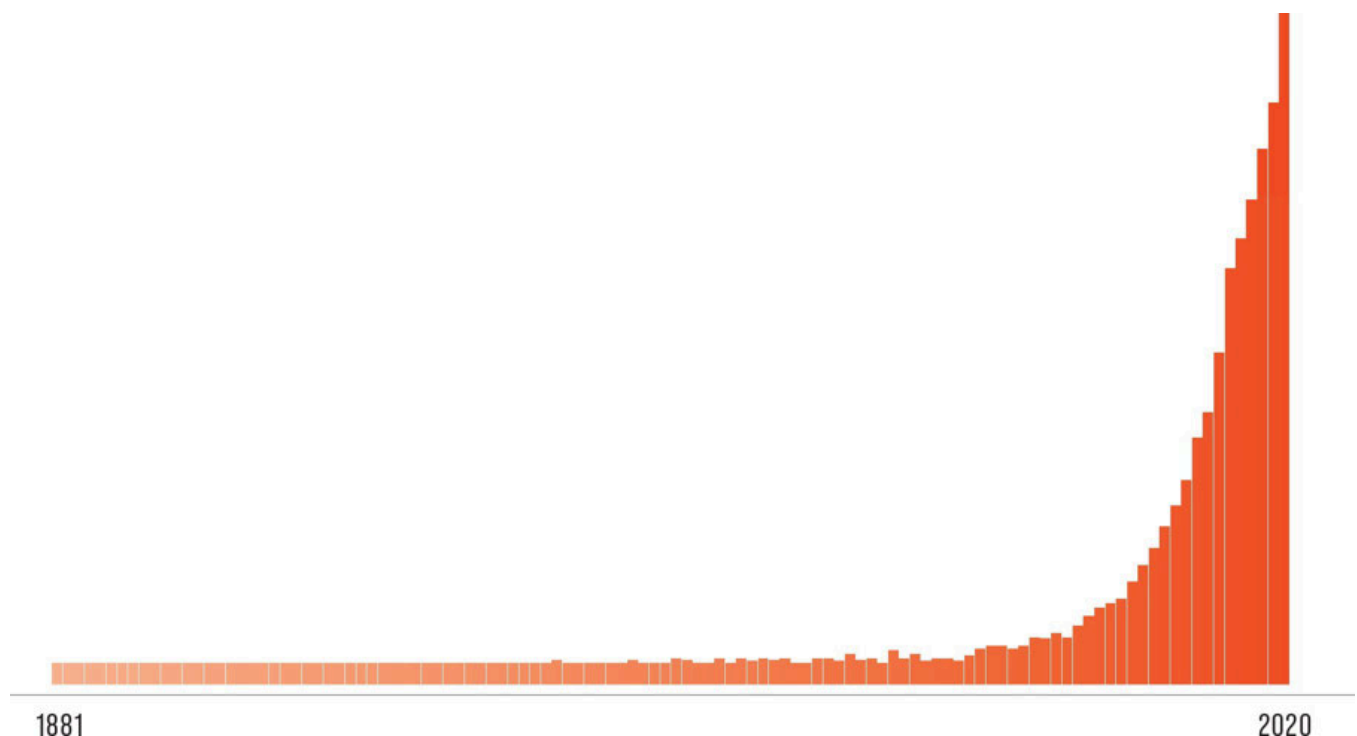


FIGURE 4.1 A PubMed search for the word “ultramarathon,” “ultra marathon,” or “ultra endurance.” Accessed March 2021.

Much of the research that has been conducted is based on pre- and postrace questionnaires asking things like: Why did you drop out of the race? What was your biggest issue? How long was your longest training run? While the answers to these questions offer a glimpse into a runner's trials and tribulations during an ultramarathon, their usefulness is limited. They are the runner's own interpretations of what happened, not necessarily clear explanations for why it happened. Take, for example, a commonly cited reason for underperformance in an ultramarathon: nausea. “I had a queasy stomach” and “I couldn't tolerate any food” are certainly important sentiments to capture, but then what do you do about them? Very little research exists into why that nausea happened to that particular individual in the first place in an ultramarathon setting. Did she take in too much food? Not enough fluids? Too much of a particular type of carbohydrate? Was the gut actually damaged and leaking endotoxins into the bloodstream? Did her vision become altered late in the race, causing disequilibrium and nausea? We know that nausea is in fact an issue, but no one knows for sure what the key triggers for nausea are in an ultramarathon setting. Make no mistake: we're getting closer and closer to finding helpful answers for this and many other questions. But you would be hard-pressed to find a singular “aha” discovery that would prevent nausea from happening in every case.

So, what do we really know about the science of ultrarunning? In 2011, Martin Hoffman and Kevin Fogard published an article titled, “Factors Related to Successful Completion of a 161-km Ultramarathon.” Their study explored the characteristics and issues that affected the performance of runners during the 2009 Western States 100 and 2009 Vermont 100 via pre- and postrace questionnaires. One of the more interesting tables in their article outlined the main problems self-reported by both finishers and nonfinishers (Table 4.1).

Through this lens, we can look at many of the failure points and limiting factors of performance and what the science has to say about them. Nausea, blisters, exhaustion, and muscle pain top the list of ailments runners

mentioned as performance limiters. Surprisingly, being inadequately trained represented only 13 to 15 percent of the complaints among both groups as a reason for diminished performance (Table 4.1) and was the least-cited reason (at 0.7 percent) for dropping out among non-finishers (Table 4.2). It would be easy to dismiss this data point, but for me it is crucial.

PROBLEM	FINISHERS (%)	NONFINISHERS (%)
Blisters or “hot spots” on feet	40.1	17.3
Nausea and/or vomiting	36.8	39.6
Muscle pain	36.5	20.1
Exhaustion	23.1	13.7
Inadequately heat acclimatized	21.0	28.1
Inadequately trained	13.5	15.1
Muscle cramping	11.4	15.8
Injury during the race	9.0	10.1
Ongoing injury	7.5	15.8
Illness before the race	6.0	5.0
Started out too fast	5.1	6.5
Vision problems	2.1	3.6
Difficulty making cutoff times	1.8	27.3
Other, not categorized	11.7	26.6

TABLE 4.1 Comparisons of problems that impacted race performance.

PROBLEM	%
Nausea and/or vomiting	23.0
Unable to make cutoff times	18.7
Other, not categorized	12.2
Ongoing injury	7.9
Injury during the race	7.2
Inadequately heat acclimatized	7.2
Blisters or “hot spots” on feet	5.8
Muscle cramping	5
Muscle pain	4.3
Exhaustion	3.6
Illness before the race	2.9
Vision problems	0.7
Started out too fast	0.7
Inadequately trained	0.7

TABLE 4.2 Main reasons given by nonfinishers for dropping out. Source: Hoffman and Fogard 2011.

I would argue that if you are nauseated, are unable to make the cutoff times, and have muscle pain that forces you to drop out or causes you significant issues, above all else, you are inadequately trained. A successful training process for an ultramarathon addresses all of those issues. Any ultramarathon will still be hard, and even the most well-prepared ultramarathon runners encounter issues, but training either mitigates or alleviates them altogether. The striking fact is that while runners often are able to identify the acute causes of their discomfort, they usually do not correlate those sensations with being inadequately trained. This is important because it means runners are focusing on the symptoms and not the root cause of the problem: inadequate training.

Using nausea as an example, you absolutely can and should train to have a stronger stomach (and I'm not talking about six-pack abs). Your digestive system is a combination of muscular and cellular machinery, and it adapts to stress just like the heart and skeletal muscles. That being the case, you can train your gut and digestive machinery to absorb calories from carbohydrate faster, to digest food more rapidly, and to resist damage caused by bacteria (Carrio et al. 1989; Costa et al. 2017; Cox et al. 2010; Harris, Lindeman, and Martin 1991; Jeukendrup and McLaughlin 2011). Furthermore, as part of the training process, you can and should use different types of calories to see what does and does not work for you.

Nausea is particularly accentuated in ultramarathon running because the event is long enough for consistently poor nutritional choices to significantly affect performance. You can eat poorly and hang on long enough to finish a four-hour marathon or maybe even a 50K. But add another twenty hours of running, and those mistakes will catch up to you. Training for the gastrointestinal stress of racing will produce positive gut adaptations, just as interval work will improve your cardiovascular system. If you are too nauseated to keep running, quite frankly, you are inadequately trained for that stressor of an ultramarathon!

Blisters, dead legs, muscle pain, and all the remaining items on the list in Table 4.2 are similar. They are all stressors in an ultramarathon, and they can all be trained. The key is that you have to know what the science says in these areas in order to know how to train them. So, using Hoffman and Fogard's list, let's take a look at these limiting factors for performance, what the science says about them, and how to properly train for them.

Limiting Factors

"What does success look like to you?" I routinely ask this of all of my athletes, whether they are just starting out, attempting to finish their first ultra, or trying to win races. This question allows me to get a comprehensive view of what they are trying to accomplish. Inevitably an athlete's answer includes some outcome goals ("I want to finish Leadville in under thirty hours") and some process goals ("I want to be able to actually run at the end of the race"). As a coach I learn a whole lot more from "I want to be able to run at the end of a 100-miler" than from "I want to finish the Leadville Trail 100 in under thirty hours." Why? Because when it comes to the outcome goal, determining the range of performances of which an athlete is capable is a matter of simple math. How they become the most adept at achieving that goal is a much bigger—and more vexing—question. By asking what success looks like, I get the color and context of the entire athlete, not just the end goal.

In my experience, most ultramarathon athletes, even the elites, find success through a lack of failure on race day. They achieve their goals, win races, and get those coveted belt buckles not because they ran one section very well but because they prevented the negative. They prevented time spent at 0 mph. They prevented themselves from becoming a nauseated, sore, blistered, battered, and stumbling mess. They continued to be able to eat, drink, and locomote down the trail, even if it was not very fast. Because so many things can go wrong and the penalties for failure are high, "success by lack of failure" is a key element in successful ultramarathon running. These failure points are somewhat universal, as indicated by Hoffman and Fogard's research, and help define the limiting factors for ultrarunning. (The very small exception is elite athletes competing in 50Ks, 50-mile distances, and flat

100Ks. This is because the finish line for elite athletes in those races often comes before the failure points discussed in this chapter have a chance to impact performance.)

All reasonably healthy individuals can locomote at the necessary speed to beat the cutoffs for any ultramarathon. I say this not as an opinion but as a biomechanical fact. The preferred walking speed for the average human is around a nineteen-minute mile (Browning and Kram 2005; Levine and Norenzayan 1999; Mohler et al. 2007). With a little effort, one can easily achieve eighteen-minute miles, which is a pace that would yield a thirty-hour 100-mile finish.

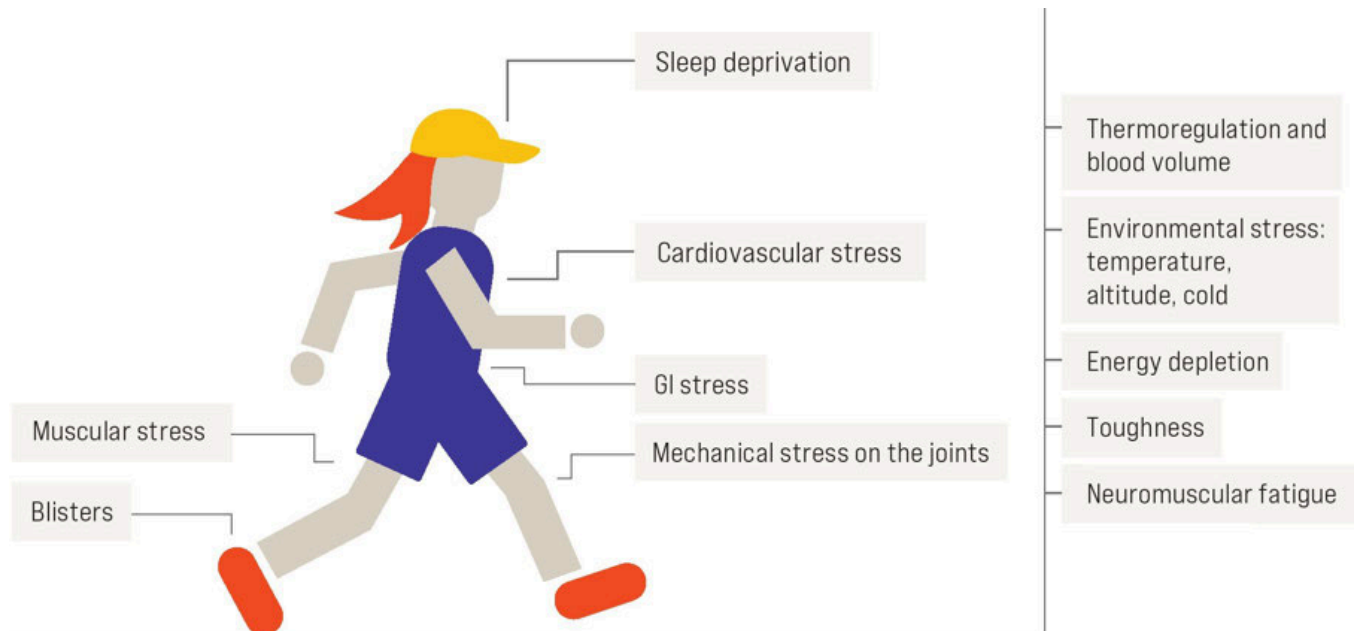


FIGURE 4.2 Ultramarathon race stressors.

Former Western States record holder Timothy Olson ran 14:46:44 in that coveted race. This time works out to about 8:50/mile. When we tested Timothy in our lab, his lactate threshold pace was under 6:00/mile. At a pace slower than 6:00/mile, his aerobic system can keep up with his energy demand, delivering oxygen to his muscles at a rate that is sustainable with few negative by-products. Having coached Timothy since he set that record, I can attest that on any given day, an 8:50/mile pace is not challenging for his cardiovascular system, even on terrain similar to Western States. Yet if you look at his Cal Street section (from mile 62 to mile 78), you will see that he ran for 16 miles with a net elevation loss at a pace of nearly 9:00/mile. From the standpoint of cardiovascular fitness, that 9:00/mile pace, which is more than 50 percent slower than his lactate threshold pace, was easy. It was essentially a normal recovery-run pace for him. So why on race day, sixty-two miles into a record-setting performance, couldn't Timothy run faster than his normal recovery-run pace? The answer lies in the fact that there are many stressors on race day, and success in an ultramarathon has far more to do with your ability to cope with the sum total of those stressors than with the capacity of your cardiovascular system alone.

Figure 4.2 illustrates the major limiting factors experienced by ultrarunners. They are not all equally limiting, but they all play some part in any ultramarathon. Your goal is to arrive at the starting line as a 100-percent-ready athlete, meaning your training has fully prepared you to handle all the potential stressors of the event.

This sounds like a simple task. However, distinguishing fact from fiction in today's world makes finding good information on how to train for these limiting factors problematic. Particularly because of the proliferation of social media and worldwide connectivity, there is more information (and misinformation) than ever about how to prepare for any endurance sport. The minute another study comes out, the popular press jumps on any new and unique angle. The 280-character version is posted on Twitter; the highlights of the article are "shared" and "liked" on Facebook; and athletes read said interpretation of the study and chase it accordingly. How can you not? You might read, "Study shows that breathing through the left nostril improves endurance performance by 12 percent" and think, Then why the heck do I need a right nostril in the first place? Some of the research is better than the

rest, some is outdated, and some is very good, but how do you wade through it all? One of the failings of popular media is that they can always find some research to support an idea on how to improve endurance performance. Additionally, like any other area of science, sports science is constantly evolving. What we knew to be true several years ago may no longer be considered true now or in the future.

With this limitation in mind, I developed my training strategy for ultramarathoners by combining research specifically done with ultramarathoners, research in other endurance disciplines that can be applied to ultrarunning, and practical coaching experience. Academics and journalists would lay out all the studies, strategies, and philosophies side by side and leave you to figure out what to do with them. That's like having a doctor hand you textbooks when you go into the office feeling like death warmed over. When you're sick, all you want to know is what to do to get better. As an athlete you want to know how to use the best that science has to offer to improve your performance. I'm going to do both—explain the science and show you how to practically use it to optimize your performance.

Hydration, Sodium, and Thermoregulation

Although an ultramarathoner will face many stressors on race day, much of the associated duress is fundamentally determined by two very simple physiological factors: hydration status and fuel availability. Of the two, hydration status is far more important. The fact of the matter is, fueling errors are easy to fix, but your hydration status is not. You can pop a gel, drink a Coke, or eat some noodles and within minutes add fuel to the fire. Your body's process for this is simple: eat, get sugar into your bloodstream, and deliver it to your muscles and brain. Even if you eat the “wrong” thing, you will still get sugar into your body relatively quickly.

If you screw up your hydration status, the fix is not so simple. Compared with fixing a bonk, the remedy involves far more complex mechanisms that include hormonal regulation, electrochemical gradients, and other ridiculously complicated-sounding phenomena. At times, staying hydrated sounds more like an engineering project than just talking about water. In addition to being more complicated than “eat sugar and let it digest,” these mechanisms of regulating hydration (and therefore blood volume) are indeed slower. They take hours to rectify if disturbed, and the series of steps an athlete may need to follow is often complicated.

More importantly, the penalty for screwing up your hydration (and therefore blood volume) is far more severe than bonking. If you run low on energy, you simply slow down and eat. At worst, you get disoriented, which in extreme cases can lead to nausea and vomiting. Don't get me wrong, those are not good outcomes, but a drop in blood volume can be much more catastrophic. I'm not trying to scare anyone off from running an ultra, but if you screw up your hydration enough, you could end up in the hospital or even die. The magnitude of the “penalty for failure” in this respect is precisely why hydration, sodium, and thermoregulation are the most important nutritional aspects to understand while preparing for an ultra.

Over the past several years, it seems like more sports science research has been done in the area of hydration and sodium consumption than in any other. This makes sense given the aforementioned importance and the overall complexity of the issue (not to mention the financial incentive for for-profit companies to demonstrate the efficacy of their drinks). In ultramarathoning, sodium supplementation has gone from being demonized to in vogue and back again. Even the most basic measurements, such as an ultrarunner's body weight (which can fluctuate based on hydration status), goes in and out of favor. As an example: for many years, high-profile races such as the Western States 100 and Wasatch Front 100 used body weight to determine if a runner was fit to continue. While that practice is no longer followed at those races, the fact that one of the simplest tools available, one that physicians have used for hundreds of years, was deemed essential and then disappeared in the matter of a few years emphasizes how complicated the practice of monitoring hydration can be.

The current battle between “drink early and often” vs “drink only in response to thirst” adds more confusion for ultrarunners. It is important to realize that in the context of hydration, water, sodium, and carbohydrate need to be considered simultaneously. These components are intertwined, with one always affecting the others. Therefore, the recommendation for when to drink always needs to begin with the answer to an earlier question: What should you drink? While some physiological aspects of fluid and sodium balance are still debatable, research has provided athletes with key recommendations for maintaining hydration and sodium balance and ultimately for

improving performance. This is such a large topic that rather than getting bogged down in it here, I have covered it in greater detail in Chapter 13.

Fueling and GI Distress

For the last several years on April Fools' Day, GU Energy Labs designed elaborate pranks that revolved around a nutrition "breakthrough." One year it was the flavor of their gels. Their lineup of "savory" flavors like Pimento Loafer, Lard Dart, and Savory Sardine was simultaneously funny and vomit-inducing. One memorable prank, though, struck a particular chord: DermaCharge, a gel that you smeared on your skin that delivered energy and electrolytes. The gag was hilarious, complete with scents of Tenacious Tomato, Sultry Cucumber, Intense Butter, and Furious Avocado, and, of course, photos of a fitness model smearing the goop on his six-pack abs. I got a good laugh out of it, but that day the questions I received by email ranged from "What do you think of this? Is it going to work?" to "How come you never mentioned this to me?" Some athletes were actually upset that they learned about this breakthrough from social media and not from their beloved coach.

At first I was stunned at how easily the wool had been pulled over their eyes. I mean, come on, a goop that you rub on your skin that smells "furiously" like avocado and delivers a wallop of carbohydrate and electrolyte? Really? Besides the lesson to be wary of marketing, their reactions reemphasized an important point: athletes are always looking for a better way to fuel. If one thing is drilled into ultramarathon runners' heads more than any other concept, it is that you must fuel in order to be successful. Even for my athletes, who have worked considerably on their nutrition strategies to minimize the stomach distress referenced in Hoffman and Fogard's research, any conceivable way to take in calories without eating sounded too good to pass up . . . even if it was as silly as rubbing avocado-scented goop on their bellies.

WHAT THE SCIENCE SAYS

At a fundamental level, digestion is a relatively simple process. When you eat a gel, cookie, or anything else at the aid station table, you must first mechanically break down the food by chewing it in your mouth, where it comes into contact with enzymes in saliva, and by churning it in your stomach. Then your stomach and intestines chemically break it down further with gastric juices, acids, and more enzymes. Finally, the intestines absorb the nutrients from the broken-down food and transfer them into the bloodstream. The key is that both a mechanical and a chemical process are in play when properly digesting and absorbing foodstuffs. Both phases of this process are central to understanding fueling and nausea.

YOUR STOMACH AND INTESTINES ARE MADE OF MUSCLE, TOO!

As I just described, your stomach and intestines are essential for digestion. Because your stomach and intestines are muscular organs, they require blood flow to do their job, just like any other muscle in your body. The problem is, blood flow is a hot commodity when you are running. At any point in time, the blood volume you have available is a fixed resource, and blood is needed to deliver oxygen and nutrients to your working muscles (including your stomach and intestines), and it also needs to move to the surface of your skin to dissipate heat. This creates a fierce competition between the aforementioned processes of digestion, oxygen and nutrient transport, and cooling. And guess which competitor wins? Ding, ding, ding! If it has to, your body prioritizes cooling, in order to protect vital organs, over delivering oxygen to working muscles (thankfully so, I might add). When this happens, blood flow to the stomach and intestines decreases, movement of food through the gut slows or stops, and pretty soon you start experiencing gastrointestinal distress. In this way, training to become more fit

increases your body's ability to process food during an event. The reason is that training increases your overall heat tolerance and reduces the blood flow required by skeletal muscles at the same running pace, both of which free up blood to be sent to the gut.

The number one recommendation when you have a queasy stomach is to "slow down and cool off." Besides being easy to remember and implement, this advice is effective because both actions redistribute blood flow from other areas back to the stomach. Slowing down reduces the oxygen and nutrient demands from your working skeletal muscles, thereby freeing up blood for digestion. Cooling off does the same by reducing your body's need to send blood to the skin in order to cool you down. You get the greatest bang for your buck by doing both because slowing down also reduces the heat generated by skeletal muscles and helps you cool down more quickly.

DAMAGE TO THE GUT

In addition to creating a competition for blood flow, endurance running causes damage to the gut as a by-product of digestion, blood flow reduction, and constant jostling up and down (Heer et al. 1987; Lucas and Schroy 1998; Øktedalen et al. 1992; Papaioannides et al. 1984). Recently researchers at Monash University studied the naturally present bacteria (endotoxins) that leak into the bloodstream as a result of this damage. They found that most individuals participating in an ultramarathon had markers in their bloodstream equivalent to those found in hospital patients experiencing an acute infectious episode (Gill et al. 2015). This means the gut is so damaged that it leaks endotoxins and triggers an immune response that is as severe as if you had a life-threatening infection. Although the researchers concluded that the damage was significant and that the gut was impaired, little evidence was presented for how to alleviate or avoid the condition. The one correlating factor suggested by the research team was that the individuals who had simply trained more exhibited less damage (Monash University 2015). Further research is needed to better understand why this extreme amount of damage to the gut occurs, but the current takeaway suggested by the research team is simple: better training equals less damage to the gut and less damage to the gut leads to less gastrointestinal distress which literally fuels better performance.

TO EAT ON THE UPHILLS OR DOWNHILLS?

WHEN AND WHAT YOU ARE DOING while you eat can affect food tolerance and calorie absorption. The less jostling you are experiencing and the more blood flow you have available, the better you will tolerate food. Furthermore, eating smaller portions of food means less blood flow is required by the gut for digestion, and you will have less stuff bouncing around in your stomach that could potentially damage it. So, when considering your next cookie, pretzel, or gel, think first about the ideal time to eat it. If you do have to take in a bigger caloric punch, do it when you have the greatest blood flow available and the least jostling—in other words, during a slower uphill hike. The goal is to have as much of that food as possible digested and into your bloodstream by the time you return to higher intensity or more jostling. Depending on the length of the climb, try to finish eating at least ten minutes before the summit. Taking in your bigger portions of calories during the slower, less intense sections of any ultra may help stave off the gut distress that can be caused by reduced blood flow and damage to the gut.

Blisters

My very first ultramarathon experience was as a crew member for Dean Karnazes during the Badwater Ultramarathon. Talk about jumping into the deep end. I was extremely nervous and legitimately underqualified. Assisting a well-known athlete in a big, demanding race in a completely foreign environment was way over my head. Therefore, before the race, I did copious amounts of research on the course, what to expect, the limiting factors for performance, and how the race worked (Badwater is unique in that the crew can accompany the runner nearly the entire way). Through that research, two things stuck out: Death Valley would be hot as hell (duh), and

runners often end up with mangled feet. I immediately made three purchases: an ice bandana, a prepackaged blister kit, and the book *Fixing Your Feet* by John Vonhof. When I opened up the package and started rifling through the contents of the blister kit, I had no clue how all the powders, lubricants, tapes, adhesives, and bandages worked. So, I read, and I practiced. I would read through a section of my new book, contort my legs like a tenth-degree yogi to gain access to my foot, apply some concoction of adhesive and tape, and then go run to test out the technique. Every day, I tried something different. As the race neared, I was obsessed. To put some of these newfound techniques to the ultimate test, and to mirror the conditions of the race—specifically the condition of constantly pouring cold water over yourself and your runner—I routinely soaked my laced-up feet with the garden hose before setting out on a run. Though this process of trial and error, I was able to refine my technique and tools and find what combinations would work for my feet in those conditions (which in this case was a combination of KT tape and tincture of benzoate).

As it turned out, I got to use my newfound skills in the wee hours of the morning in Death Valley, as Dean managed to get a blister smack-dab on the ball of his right foot. Somehow my tape job held up for the remaining forty miles of the race. After Dean's race and our celebratory dinner were over, we staggered into the Dow Villa Motel in Lone Pine, California. The motel is on the racecourse (mile 122) and is used as a medical checkpoint and communications hub for the event. A constant stream of runners ran (or staggered) past the motel throughout the night, illuminated by a union of streetlights, headlamps, multicolored safety lights, and reflective vests. Curious and hopped up on far too much caffeine, I stayed awake all night watching the battered runners gradually make their way down the course. Most of them looked awful, limping and moving very slowly through the darkness. They were dealing with a variety of issues: tired legs, fried brains, hyponatremia, bonking, hallucinations, you name it. However, one prevailing issue bound them together as brothers and sisters of this race: collectively, their feet were destroyed. Runner after runner stopped at the motel to get their feet patched up, sometimes by professionals and sometimes by their crews. Some foot issues were relatively benign. Most were dreadful. All night and into the next day, the runners came in, got patched up, and left ready to tackle the final thirteen miles of the race.

Upon returning home, I decided to reinforce my blister prevention and treatment arsenal and purchased a red plastic toolbox from Home Depot. I filled the box with an array of products to help repair feet when they become battered, bruised, bloodied, and blistered. Admittedly, the toolbox was a bit makeshift—its contents were somewhere between the basic necessities and a full-blown medical kit—but it was mine, and I knew how to use every product in it. More than fifteen years later, I still have this same box. I bring it with me whenever I attend an ultramarathon as a coach or as an athlete. Often it remains idle. At other races it is a lifeline for athletes. I take comfort in knowing that if athletes do run into trouble with their feet, I have some level of skill to patch them up and get them on their way.

Hoffman and Fogard's survey of participants in the Western States 100 and Vermont 100 demonstrates that blisters remain a prevalent issue and do limit performance. Quite simply, your feet propel you forward. When your foot is damaged, it affects the entire kinetic chain from the ankle to the knee and through the hip. You might be able to limp through for a while on a sore foot, but chances are that the change in biomechanics will eventually catch up with you and compound the issues you face. In many cases, it's not the blister that leads to the DNF. The blister just starts the process by changing the way you run, and over time those changes lead to other biomechanical problems that slow you down, knocking you off your nutritional strategy, exposing you to the elements longer, and so forth. This does not have to be the case. Blisters can be largely prevented through training, prevention, a little treatment know-how, and better race-day management. Treatment is relatively easy, requiring a few basic products and skills to fix the majority of issues. It takes practice, but it's worth it.

PREVENTING AND TREATING BLISTERS

Whenever you stress an organ or a structure in your body beyond its capabilities, you cause damage. Ultramarathons normally represent a longer, more difficult run than your day-to-day training, complicated by the fact most ultramarathon events occur in areas away from your home training grounds. The trail surface, camber, dirt, dust, and debris your feet encounter are undoubtedly different during the race than at home. Furthermore, your biomechanics are different depending on the properties of the trails, placing stresses on different areas of the foot—and, of course, the skin covering those areas. Consequently, the shoe/sock/powder/tape/lubricant/insole

combination that worked in training may not always work during the race. Just as training on flat ground will not completely prepare you for a mountainous ultra, training on your home trails might not fully prepare your feet for the rigors of race day. Therefore, a combination of education, preventive measures, and wound care skills offers the most comprehensive way to ensure that your hard-earned training is not negated by the unraveling of your feet on race day.

WHAT THE SCIENCE SAYS

What runners commonly refer to as a blister is clinically termed a “friction blister” because friction is the primary culprit. Although heat and moisture are contributing factors, friction and the underlying shear forces (forces parallel to the ground) are what ultimately cause the dreaded blister (Figure 4.3). As I will discuss later in this chapter, with each and every foot strike, you apply shear forces parallel to the surface of the ground in both the anterior-posterior (forward-backward) and the mediolateral (side-to-side) directions. As you hit the ground, the ground pushes backward on your shoe, your insole pushes backward on your sock, and your sock pushes backward on your skin. Then as you push off the ground, these forces between your skin and sock, sock and insole, and shoe and ground all reverse direction.

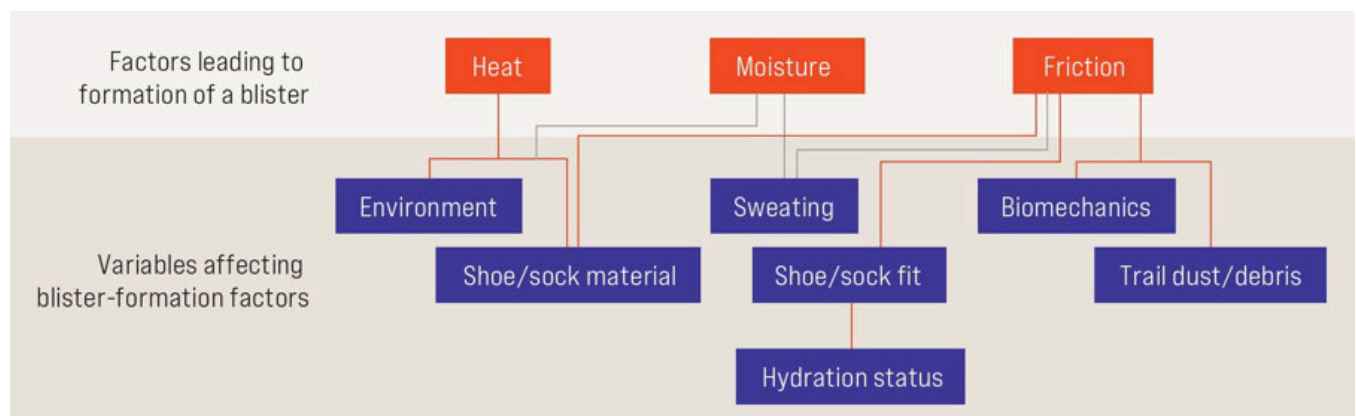


FIGURE 4.3 Heat + moisture + friction = blister.

The problem is that the surface of your skin is pliable (after all, it is called soft tissue). As your body applies these shear forces, your soft tissue (skin on the feet) moves more than your skeletal system (rigid bone). These out-of-sync movements between your skeleton, soft tissue, sock/shoe, and shoe/ground are what ultimately cause the frictional force that leads to a blister. Your shoe and sock move against your outer layer of skin (epidermis) more than your outer layer of skin moves against your inner layer of skin (dermis). As the bump and grind between these two skin layers continues, the layers eventually separate. Once this separation occurs, fluid fills the void due to hydrostatic pressure. The result is a fluid sac between the newly separated layers of skin bordered by yet-to-be separated layers along the edges.

The addition of heat and moisture exacerbates blister formation. Heat accelerates the blistering process primarily by loosening the bond between the dermis and epidermis. Research has demonstrated that an increase in skin temperature of 4°C will accelerate blister formation by 50 percent (Hashmi et al. 2013; Kiistala 1972b). Increases in foot skin temperature, heat from the environment, metabolic response, and heat from frictional forces all add to the heat within your shoes. Plus, the warmer your feet are, the more they sweat. The more your feet sweat, the more saturated your socks, shoes, and skin become. Dumping water on your head, running through a river crossing, and running in the rain all accomplish the same feat (excuse the pun): they increase the moisture in your shoe, in your sock, and on the skin of your foot. This moisture further increases the frictional forces between your foot and sock with every foot strike, adding to blister susceptibility (Naylor 1955; Worthing et al. 2017). Furthermore, the moisture weakens the outer layer of skin, making it more prone to injury.

While heat, trauma, and moisture are attacking your skin from the outside, on the inside your hydration status can also make you more susceptible to blisters. If a runner becomes even slightly hyponatremic (having low blood sodium), one of the body's protective mechanisms—long before blood sodium levels are dangerously low—is to pull water out of the plasma and into the extracellular space, thus increasing the concentration of sodium in the blood. As a simple by-product of gravity, this fluid in the extracellular space pools in the extremities, commonly resulting in puffy hands or feet. Unlike your hands, your feet are subject to the rigors of propelling you forward. The increased fluid in the extracellular space in the skin of your feet accelerates the blistering process: it causes your feet to get bigger, turning your once perfectly broken-in shoes into friction-laden traps. It also loosens the skin layers, enabling fluid to be pushed between the dermis and epidermis more easily.

TRAINING, GEAR, AND PROTECTION

Each of the three aforementioned blister factors of friction, heat, and moisture can be alleviated with the right combination (in order of priority) of training, gear (shoes and socks), race-day preventive measures, and finally treatment.

How Training Influences Blister Formation

Training is the first level of prevention in blister formation. Your skin adapts to stress just like any other organ in your body. Many studies, primarily involving the military, have demonstrated that gradual exposure to frictional forces on the foot (through hikes and marches) decreases the skin's susceptibility to blisters (Allan 1964; Hodges, DuClos, and Schnitzer 1975; Knapik et al. 1995). As you train, your epidermal skin cells become thicker and in theory more cohesive, making them more resistant to blistering. How does this happen? As you run, you slough off skin cells faster than normal. These are rapidly replaced by new skin cells, but the young cells don't get the chance to differentiate into layer-specific cells (epidermis, dermis) before they are stressed by another run (Kim et al. 2010). When this happens frequently over a relatively short time, it results in overthickened skin, otherwise known as a callus.

SHAVE DOWN YOUR CALLUSES

SHOULD YOU KEEP YOUR CALLUSES or shave them down? Proponents of keeping the callus say the extra-thick skin is less prone to damage and therefore acts as a protective layer. Although there is some truth to this, the far greater risk is that the callus will continue to grow and become an anomaly in the foot/shoe interface (i.e., it sticks out). Remember that a blister forms when an outer layer of skin moves out of sync with an inner layer of skin. A callus can act as the outer layer and still separate from the inner layer of dermis, leading to a blister under your callus! When this happens you generally lose the entire callus, which defeats the purpose of building up this protective layer of extra thick skin.

Because allowing calluses to become overly thick greatly increases the likelihood of forming a blister under the callus, I encourage athletes to keep calluses shaved down to maintain a smoother, more uniform skin surface. That doesn't mean you should remove calluses entirely. Use a pumice stone or metal callus shaver and file down the callus so that it is flush with the surrounding skin. Shaving them is a matter of maintenance, not removal.

How Shoe and Sock Choice Influences Blister Formation

Your shoe and sock combination is the next level of blister prevention. With respect to blister prevention, your shoe/sock combination should serve the dual purposes of (1) reducing frictional forces between your skin and sock and your sock and shoe and (2) managing moisture transfer from your foot into the air. Some socks that segregate the toes (toe socks) also aim to reduce the skin-to-skin friction between the toes. Overwhelming research has shown that a well-fitting wicking sock offers the best blister prevention strategy (Herring and Richie 1990; Knapik et al. 1995, 1996). Fortunately, most sock companies are now wise to this idea and are moving away from thicker, bulkier, and less wicking socks for runners and ultrarunners. Shoe companies are following suit. Materials for shoes are constantly becoming more pliant and breathable, and shoes are available in a wider variety of shapes and sizes. Perhaps blister prevention is not the shoe manufacturers' end goal, but the improvements in materials and fit do help.

You should test your shoe/sock combination in training and conduct shoe trials early in the season. It is important to figure out what works and what doesn't far out from your event so you can ramp up your training without fear of damaging your feet. Along these lines, once you find the combination that works for you, I recommend investing in enough shoes and socks to get you through the entire season. Ultrarunners burn through shoes and socks, and designs or availability can change unexpectedly. You don't want to be searching for new shoes or having to switch to a new model or design in the middle of the season when your training workload is very high.

Race-Day Preventive Measures

One lesson I've learned from racing and crewing ultramarathons—and from watching that parade of mangled feet at Badwater—is that athletes use an incredibly wide range of techniques to prevent blisters on race day. Some use tape, some use lubricants, and some use powders. The more creative ultrarunners will use elaborate combinations and concoctions of all of the above, sometimes taking hours to apply. In many cases, two runners will use techniques with opposing goals (keep the skin from moving vs encourage the skin to move with less friction), and both techniques may (or may not) work. This reinforces the “find what works for you” advice from the bible of blister prevention and treatment, *Fixing Your Feet*, which makes the point by providing numerous personal “n of 1” anecdotes from athletes. From a scientific standpoint, this advice rings true. In fact, there is little scientific evidence that any of the aforementioned strategies work. Furthermore, there is conflicting research indicating that some preventive measures actually exacerbate the problem by adding moisture and thus increasing skin friction (Figure 4.4; Allan 1964; Allan and Macmillan 1963; Knapik, Reynolds, and Barson 1998; Knapik et al. 1995; Nacht et al. 1981; Quinn 1967; Reynolds et al. 1995).

This is an area where less is more. The prevention strategies we know work are training (conditioning the feet to handle the stresses of long miles) and an effective shoe/sock combination. These should always be your starting points. Adding other techniques like tape, lubricant, powder, or antiperspirant increases complexity, adds more variables to the training or race-day situation, and might exacerbate the problem.

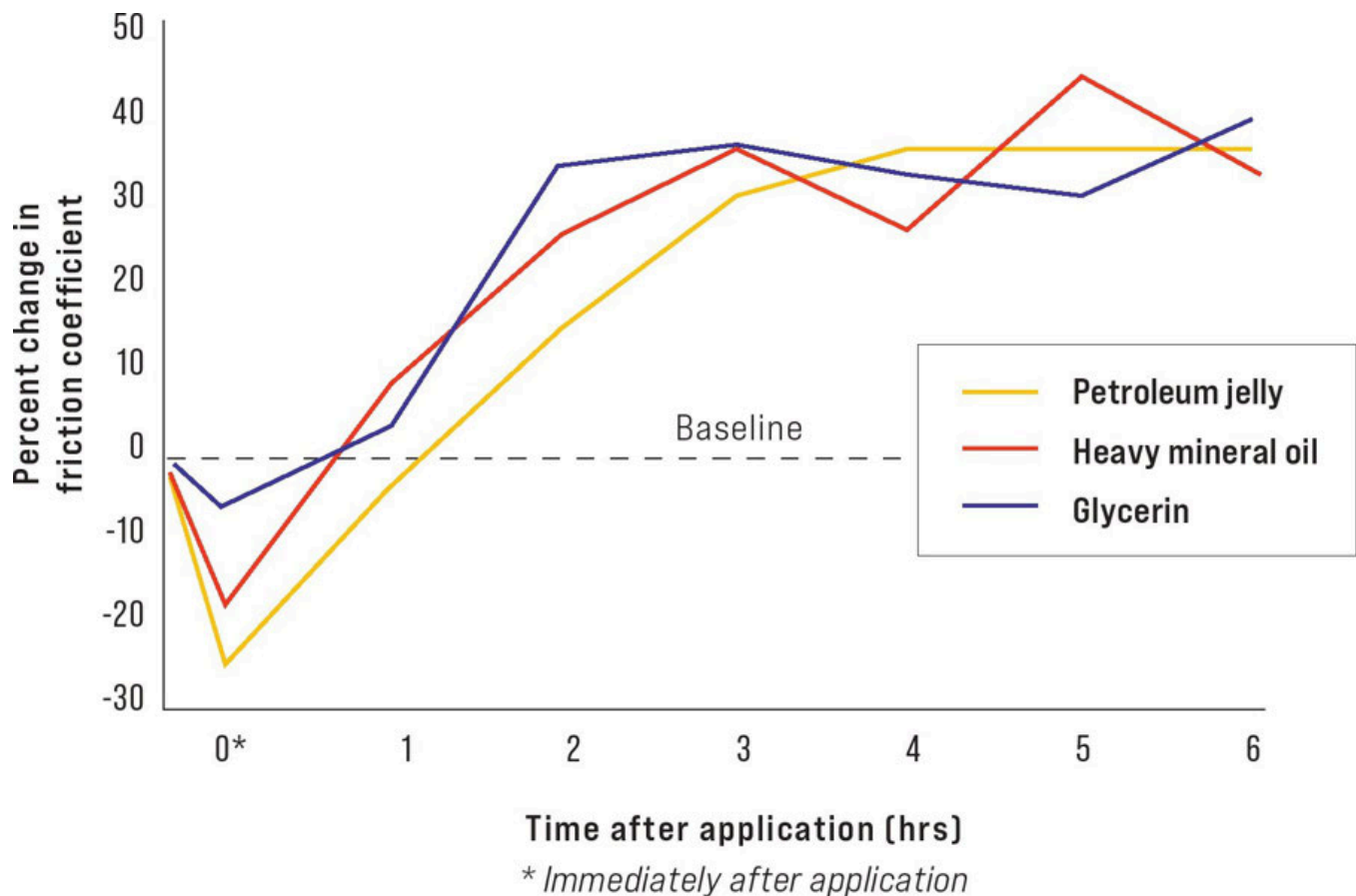


FIGURE 4.4 Graph showing an initial decrease, then increase, in friction of common lubricants when used on the skin. Source: Nacht et al. 1981.

It is also important to realize there is no combination of equipment or amount of training that will entirely eliminate the risk of developing blisters during training or on race day. You can minimize these risks as much as possible, but you also have to learn to patch and repair a blister in the field when things go wrong.

Blister Patch and Repair

If you do get a blister (or the precursor, which is referred to as a “hot spot”), you have a decision to make: you can save some time and continue running, or you can stop and lose some time treating your feet. In making this decision, you need to balance your race-day goals, performance expectations, safety, and race situation. Generally speaking, the more time you have left to run and the bigger the problem could become, the more it is worth your while to take a few minutes and fix what is wrong. Don’t let little problems become big problems. My advice is to always err on the side of caution and fix problems early, particularly at the 100K and 100-mile distances, where there is a lot of ground to cover.

If you are in a situation where you choose not to stop and fix a blister, or you are many miles from the next aid station and you have no products to treat the blister, it is time to suck it up. Blisters hurt because the foot is highly innervated, and runners tend to find relief by changing their gait or foot-strike pattern. While this is a logical strategy (“I have pain there; I am going to try to avoid it”), the ramifications of changing your gait too much can have consequences up the kinetic chain. Your foot, ankle, knee, and hip are all connected and constantly affect one another. Although I am an advocate for manipulating gait and biomechanics in an effort to combat muscular fatigue (as you will see in the next section), I do not advocate doing so in the context of a blister. In this case it’s time to be tough and keep your gait as normal as possible. Running with your normal gait may make the blister

worse, but that's still only one problem and one you can treat and get under control. Changing your gait to "run around a blister" can lead to pain or injuries you can't effectively treat and control while continuing to race. Of course, while you might have to tough it out and run on a blister on race day, it's not a strategy to rely on. Blisters can and should be prevented and treated.

WHAT A BLISTER KIT SHOULD CONTAIN

THIS SMALL ASSORTMENT OF PRODUCTS will be enough to fix minor and moderate blisters out in the field and keep you moving. It is manageable for your crew to carry or to pack in a drop bag. It is neither a substitute for a full medical kit nor what you would use to treat skin injuries after a race.

- **Adhesive felt sheet or moleskin**
- **Needles or small scalpel (size 11)**
- **Alcohol pads or Betadine swabs**
- **Gloves**
- **Kinesio Tex Gold tape, Elastikon tape, and/or Leukotape (to patch or prevent)**
- **Scissors**
- **Adhesive such as tincture of benzoin**
- **Lubricant such as Body Glide, BlisterShield, or Squirrel's Nut Butter**
- **Gauze pads**

Blisters come in a variety of shapes, sizes, and levels of discomfort. Treatments also come in many shapes and forms. Unless you are a medical professional with many years of blister management experience, a simple solution is always best. I have found success with the following nine-step plan:

1. Clean the surface of the blister and the surrounding skin. If an alcohol pad or disinfectant is available, use it. If not, it is still usually best to proceed to step 2. You are less prone to infection if you can properly manage the blister while it is small and treatable. Large broken blisters will become prone to infection more readily than small broken blisters because there is more opportunity to become infected through the larger area of damaged and exposed skin.

2. Puncture the blister with a sterile needle, sharp scissors, or scalpel. (I prefer the scalpel, as you can use it to create a window or hole for the blister to drain better.) Take care to puncture the blister enough to allow fluid to drain but not so much that the blister roof becomes detached. If you are using a needle (sterilized safety pins from a race bib also work well), put three to four holes in the blister so that it will drain. Ideally, position the punctures such that fluid can continue to drain while you keep running.

3. Squeeze the fluid out of the blister.

4. Clean and dry the surface of the blister and the surrounding skin. You are now prepping the skin to apply a patch, so ensure that it is dry and free of debris. You can choose to add a very small dab of lubricant to the blister roof. This is to prevent the patch from sticking to the blister roof when you eventually peel the tape off.

5. Size up the area you are going to dress and cut a piece of tape or bandage to cover the blister. It should be large enough so that it can stick to the surrounding undamaged skin. If the blister is on a toe, this might mean wrapping the entire toe. If you do have to wrap a toe, it's usually best to wrap the adjacent toes as well so that the tape does not rub directly on adjacent skin.

6. Apply a tape adhesive such as tincture of benzoin to the area surrounding the blister. Although the tape has its own adhesive backing, using an additional tape adhesive will ensure a better stick.

7. Place the tape down on the skin from one edge of the tape to the other. Be careful to avoid folds and creases. If you do get a fold or a crease, start over.

8. Lightly press down on the patch to ensure the adhesive completely sticks to the skin.

9. Put your socks on, lace up your shoes, and run!

If you are particularly blister prone, practice various techniques at home. Cutting and placing the patch on the surface of the skin can be the most frustrating part of the process during a race. The tape is sticky and adheres to itself and to your fingers. You're in a hurry. You're sweaty and dirty. And you're working in a dirty, dusty environment. Finding a routine and learning some simple skills goes a long way toward making the process smoother and faster in race conditions. As with any other skill, practice makes perfect!

Muscular Breakdown and Fatigue

The overall concept of muscular breakdown and fatigue incorporates several areas of physiology. Physical trauma to the muscle, depleted energy stores, neuromuscular dysfunction, central vs peripheral fatigue, and myriad other phenomena fall under this umbrella. To avoid recreating a muscle physiology textbook, I will focus specifically on the aspects of ultramarathoning that lead to muscular breakdown and fatigue.

ULTRAMARATHONS AND MUSCULAR BREAKDOWN

It is well documented that a significant amount of muscular breakdown occurs during an ultramarathon. Researchers have studied the blood parameters that indicate muscular trauma, particularly creatine kinase (CK), from finishers in Spartathlon, the Ultra-Trail du Mont-Blanc, and the Western States 100, among others. With respect to muscular breakdown, all the research comes to the same conclusion: there's a lot of it in ultrarunning. Additionally, there's tremendous variability among individuals. Blood markers for CK after an ultramarathon range from relatively normal to more than 100 times normal values (Fallon et al. 1999; Kim, Lee, and Kim 2007; Millet et al. 2011; Overgaard et al. 2002). That's a lot of muscular tissue turnover. This is all obvious to ultrarunners, even if they've never heard of CK or any of the other biochemical markers associated with pain, fatigue, and muscular breakdown, because they run slower at the end of a race.

Although it's clear that muscular breakdown and fatigue are significant, there's so much individual variability that it is difficult to determine ways to prevent them (Figure 4.5). Correlations between muscular fatigue and training components such as volume and vertical gain/loss are scant. The same goes for correlations between any race-day phenomenon (nutrition, pacing, hydration) and muscular fatigue or damage. Whenever this is the case, it is important to rely on best practices and research in other areas to help guide training principles.

With that as an introduction, a quick biomechanics lesson is in order.

During the course of running, your muscles can only pull, or contract, to move your limbs. They cannot push. When muscles pull, you generate positive force. This positive force is what pushes you off the ground and propels you forward. However, during the course of a step on level ground, you need to generate an equal amount of negative force (or counterproductive force) as your foot initially hits the ground and your body is lowered. During this initial phase of foot strike, many of your muscles are actively being lengthened, or pulling against a force in the opposite direction. Make no mistake: they are still pulling; they are just doing so against a larger force and are therefore lengthened. This active lengthening is what is oxymoronically referred to as an eccentric contraction. Such contractions are unavoidable, whether one is running or walking (DeVita, Helseth, and Hortobagyi 2007; Enoka 2008).

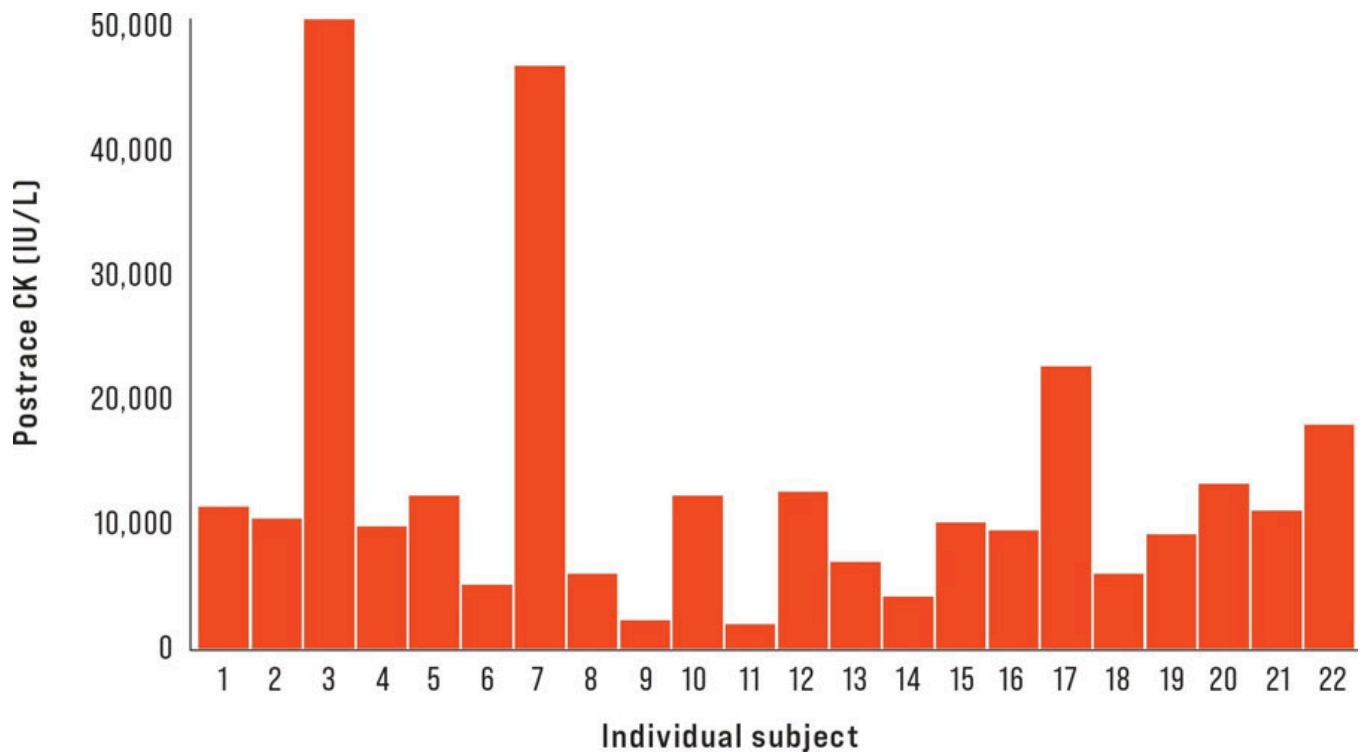


FIGURE 4.5 Postrace creatine kinase (CK) levels in Ultra-Trail du Mont-Blanc finishers. Adapted from Millet et al. 2011.

STEP VS STRIDE

FOR YEARS THE WORDS “STEP” AND “STRIDE” have been used interchangeably (and thus incorrectly) in popular literature. For the purposes of this book and to be accurate, I am going to use these terms correctly. A step is defined as one foot strike to the opposite foot strike, in terms of either length or time (i.e., the distance between your left foot hitting the ground and your right foot hitting the ground, or the time that passes between the two footfalls). A stride is defined as one foot strike to the same foot striking the ground (i.e., the distance or time from your left foot hitting the ground to your left foot hitting the ground again). Therefore, your stride length will be double your step length, because two steps essentially equal one stride.

- **Step length:** the distance between the initial contact of one foot and the initial contact of the opposite foot
- **Step rate:** the total number of right and left foot strikes per minute
- **Stride length:** the distance between the initial contact of one foot and the next initial contact of the same foot
- **Stride frequency:** the total number of the same foot strikes per minute (e.g., the total number of right foot strikes per minute)

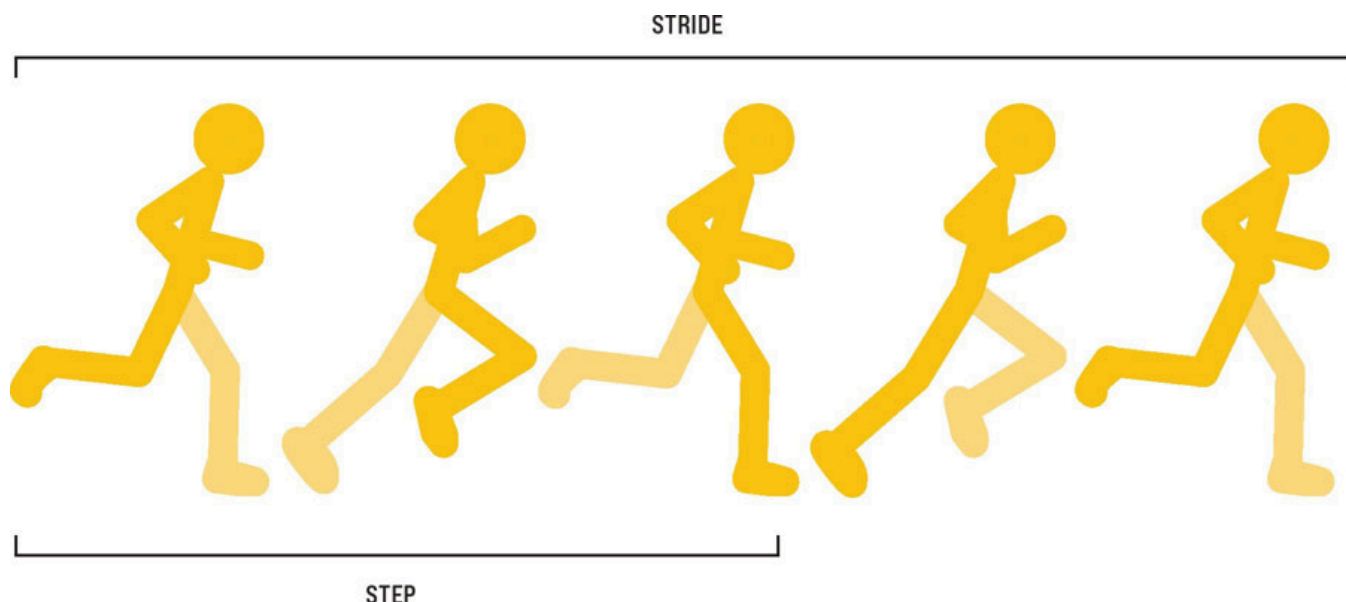


FIGURE 4.6

Whenever an eccentric contraction happens, a certain amount of muscular breakdown occurs. The amount of breakdown varies with the individual and with velocity, force, and the total repetitions that are produced (Chapman et al. 2006; Nosaka and Clarkson 1996; Tiidus and Ianuzzo 1982)—all of which is fine and dandy to know, but what do you do about it?

HOW TO COMBAT MUSCULAR BREAKDOWN

Eccentric contractions and the muscular breakdown associated with them have long been villains in the coaching and training world. The victims are your muscles, which lose their ability to function when you have stronger, more frequent, and a higher total number of eccentric contractions (Bontemps et al. 2020; Chapman et al. 2006; Ebbeling and Clarkson 1989; Eston, Mickleborough, and Baltzopoulos 1995; Proske and Allen 2005; Proske and Morgan 2001). By its very nature, an ultramarathon creates these conditions because of the hundreds of thousands of steps and associated eccentric contractions required to reach the finish line. Adapting your physiology and finding strategies to handle this huge stress can be a big advantage. Failing to do so can result in a catastrophe—so much so that ultrarunners have contrived numerous techniques to solve the problem. Equipment such as trekking poles and highly cushioned shoes have been used in attempts to minimize the damage. Additionally, training techniques such as strength training and hard downhill repeats have also been utilized by ultrarunners to combat this phenomenon. While many of these efforts deserve credit for trying, problems arise when you look at the science and the practicality of dedicating your precious time and effort specifically to some of these interventions.

While the science is not crystal clear on this aspect of performance, most research points to the rationale that a little eccentric training will go a long way in preventing muscular breakdown and that those adaptations last a long time (Clarkson et al. 1992; Nosaka et al. 2001). What is not clear is what the curve of diminishing returns looks like in this area and how we should determine the optimal amount of eccentric training.

ARE WOMEN MORE FATIGUE-RESISTANT THAN MEN?

EVERY YEAR, SOME ULTRAMARATHONS are won outright by women, leading certain people to ask, “Are women better at ultradistance events than men?” Adding fuel to the fire are anecdotal tales from expedition-length adventure racers about the women commonly outperforming the men on mixed teams late in the race. Many in the ultrarunning world have speculated that women are more psychologically and physically suited for ultramarathons than men, and scientists and researchers have debated whether women should or should not be able to outperform men in ultrarunning, noting the statistical performance differences between men and women at different distances. Early in this statistical debate, it was theorized that women could, in fact, outperform men as the distance increased (Bam et al. 1997). Later statistical analyses, however, contradicted those initial theories (Peter et al. 2014; Zingg et al. 2014). Bringing this argument full circle, recent research into the difference in fatigue resistance between men and women might now actually begin to turn the initial statistical speculations into physiological reality.

Researchers examined different variables related to fatigue for male and female finishers of the 2012 version of the Ultra-Trail du Mont-Blanc (that year’s race was shortened to 100K due to inclement weather). What did they find? While both men and women demonstrated similar amounts of central fatigue and muscular damage and inflammation (CK, C-reactive protein, and myoglobin levels), they differed in the amount of peripheral fatigue (fatigue within the muscle itself) measured after the race. The 100K race negatively affected men more than women specifically in the amount of voluntary force produced in the knee extensor and the evoked mechanical response in the plantar flexor (Temesi et al. 2015). Simply stated, within these muscle groups, women fatigued less than men after 100K of mountainous running. Does that translate to women being better ultramarathon runners than men? At this point, no. But the door is certainly open!

SHOULD I CHANGE MY STRIDE?

RECENT RESEARCH HAS SUGGESTED that one way to cope with the massive amount of muscular breakdown is to vary your stride throughout the course of the race (Bontemps et al. 2020; Rowlands et al. 2001). If you are a heel-striker, you would intentionally run with a forefoot strike periodically and vice versa. The theory is that the slight changes in biomechanical patterns will spread the work and subsequent damage out among different muscle groups. In trail running, with its varied terrain, this happens to a certain extent naturally. But in events where the terrain is more homogeneous, consciously altering your gait could prove to be a useful tool to help prevent muscular breakdown.

Additionally, some academic research has suggested that downhill running prowess has more to do with movement coordination than with oxygen consumption or force tolerance (Minetti et al. 2002). Thus, training to run downhills harder (at higher cardiovascular intensity levels and higher forces) might not have as much of an effect as training to run downhills with better technique.

In contrast, we know a lot more about the recovery from large bouts of eccentric training and associated muscular breakdown, and we can use that information to evaluate and compare eccentric training with other training components. The more you descend, the longer recovery times you will need. Additionally, the faster you descend (the higher forces you apply), the longer recovery times you will need. Comparing foot for foot of elevation change, it will take longer to recover from descending than from ascending, even though ascending requires more energy to perform and a longer time to complete.

RESEARCHERS WANT YOU!

THERE IS A LACK OF COMPREHENSIVE RESEARCH in ultramarathon running. While there are a few dedicated research teams delving into the science of ultrarunning, they are in a vertical kilometer’s worth of an uphill battle finding subjects to poke and prod. This does not have to be the case! You can help. I encourage you to participate in the research being done in ultramarathon running. Some of this is done at races, and some is conducted in the lab. If you come across a researcher asking for subjects, please consider participating. Your involvement increases our understanding of the discipline. It might even help to save a life.

Practically, we need to look at this from a cost-benefit ratio standpoint. It doesn't take a lot of descending to achieve the desired adaptation, and more descending requires disproportionately long recovery periods. It is for these reasons (and a couple more I will explain in the next chapter) that my philosophy for coping with muscular fatigue and breakdown revolves around a singular strategy: match the grade and speed of your training to what you will experience during the race. This concept is examined further in Chapter 5.

KEY POINTS ON LIMITING FACTORS

We are learning more and more each year about ultramarathon performance. We can now lean on practice and science to drive training.

Ultramarathon stressors are not the same as marathon or other traditional endurance stressors. Performance will be dictated by more than one's VO2 max, lactate threshold, and the cost of running.

Ultramarathon performance can be limited by interrelated factors such as hydration, thermoregulation, muscular damage, and gut function.

Any ultramarathon limiting factor can be trained! That's what this book is for.



